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▶ A system is in equilibrium if and only if $F \equiv 0$

► $\langle U \rangle$ is the total change in energy of the system added (called the change in enthalpy)

► $T_S < 0$ Is the energy in the system attributed to random fluctuations



By the definition of entropy we have

$$S(\beta) \equiv \beta \langle U \rangle_{\beta} + \log Z(\beta)$$

A = U - TS

▶ Hence, we have the Gibbs relation in chemistry

Define the Helmholtz-free energy as

$$F(\beta) = -\frac{1}{\beta} \log Z(\beta)$$

► In physics a reaction is spontaneous if and only if $\Delta F < 0$

FREE ENERGY

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- ▶ By the definition of entropy we have

$$S(\beta) = \beta \langle U \rangle_\beta + \log Z(\beta)$$

- ▶ Define the Helmholtz-free energy as

$$F(\beta) = -\frac{1}{\beta} \log Z(\beta)$$

- ▶ Hence, we have the Gibbs relation in chemistry

$$F = \langle U \rangle - TS$$

- ▶ In physics a reaction is spontaneous if and only if $F < 0$
 - ▶ $\langle U \rangle$ is the total change in energy of the system added (called the change in enthalpy)
 - ▶ $TS < 0$ is the energy in the system attributed to random fluctuations
- ▶ A system is in equilibrium if and only if $F = 0$

EXAMPLE: GAS IN A BOX

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